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Battery powered compact wire feeder

Abstract

A wire feeder that is powered at least partially by a battery is provided. The wire feeder includes a welding wire spool and a welding wire drive system for receiving welding wire from the spool and feeding the wire through a flexible conduit attached to an enclosure of the wire feeder. A motor of the wire feeder drives the welding wire drive system, powered at least in part by the battery. The wire feeder includes a battery receptacle for receiving the battery. In certain embodiments, the wire feeder may be attached to a body of a user via a harness system.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to, and is a continuation of, U.S. Nonprovisional patent application Ser. No. 13/427,353, filed on Mar. 22, 2012, which claims priority to U.S. Provisional Patent Application No. 61/467,437, filed on Mar. 25, 2011, both of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) This invention relates generally to wire feeders for welding systems. More specifically, this invention relates to a compact, battery powered wire feeder.

(2) Welding systems generally include an electrode configured to pass an arc between a torch and a work piece, thereby heating the work piece to create a weld. In many systems, such as metal inert gas (MIG) welding and stick welding systems, the electrode is a consumable wire that melts into the weld to provide a filler material into the weld. In other words, a single wire serves both as an electrode and as the filler material. In contrast, other welding systems, such as tungsten inert gas (TIG) welding systems, employ a non-consumable tungsten electrode that is independent from the filler material. In other words, the TIG welding process does not melt the tungsten electrode into the weld forming on the work piece. Instead, a user generally holds the TIG torch with the tungsten electrode in one hand, while the user simultaneously and independently holds the filler material in another hand. The independent filler material is typically provided in a 36 inch long rod, which is manually delivered to the work piece, forming a weld that is approximately twelve inches in length. When the filler rod is almost entirely consumed in the welding process, the operator stops welding, discards the remaining stub of filler material, and begins welding again with a new filler rod. Unfortunately, this process of starting, stopping, and discarding unused filler material may result in an inefficient use of time and consumables throughout the welding process. In addition, the 36 inch long rods of filler material generally used in the welding process may offer limited flexibility to the user when the work piece is located in a relatively tight space.

BRIEF DESCRIPTION

(3) In one embodiment, a welding wire feeder includes an enclosure. The enclosure includes a hub for receiving a welding wire spool and a welding wire drive system for receiving welding wire from the welding wire spool and feeding the welding wire through a flexible conduit attached to the enclosure. In addition, the enclosure includes a motor for driving the welding wire drive system and a battery receptacle. The motor is powered at least in part by a battery, and the battery receptacle is for receiving the battery.

(4) In another embodiment, a cordless welding wire feeder system includes a hub for receiving a welding wire spool and a welding wire drive system for receiving welding wire from the welding wire spool and feeding the welding wire through a flexible conduit attached to the enclosure. In addition, the enclosure includes a motor for driving the welding wire drive system and a battery receptacle. The motor is powered at least in part by a battery, and the battery receptacle is for receiving the battery. The cordless welding wire feeder system also includes a harness system for attaching the welding wire feeder to a body of a user.

(5) In another embodiment, a method for operating a welding wire feeder includes driving a welding wire drive system using a motor powered at least in part by a battery disposed in a battery receptacle of an enclosure of the welding wire feeder. The enclosure is configured to attach to a body of a user via a harness system. The method also includes feeding welding wire from a

welding wire spool disposed in the enclosure through a flexible conduit attached to the enclosure using the welding wire drive system.

Description

DRAWINGS

- (1) These and other features, aspects, and advantages of the present invention will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:
- (2) FIG. 1 is a block diagram of an embodiment of a welding system utilizing a battery powered portable welding wire feeder;
- (3) FIG. 2 is a diagrammatical representation of an embodiment of certain components of the wire feeder of FIG. 1;
- (4) FIG. 3 is a block diagram of an embodiment of certain components of the wire feeder of FIG. 2, including various control circuitry components for controlling operation of the wire feeder;
- (5) FIG. 4 is a perspective view of an embodiment of the wire feeder of FIG. 2 attached to the back of a user via a harness system such that welding wire is conveyed to the left side of the user;
- (6) FIG. 5 is a perspective view of an embodiment of the wire feeder of FIG. 2 attached to the back of a user via a harness system such that welding wire is conveyed to the right side of the user;
- (7) FIG. 6 is a perspective view of an embodiment of the wire feeder of FIG. 2 attached to the back of a user via a harness system such that welding wire is conveyed upward from the wire feeder;
- (8) FIG. 7 is a perspective view of an embodiment of the wire feeder of FIG. 2 attached to the front of a user via a harness system such that the welding wire is conveyed to the right side of the user;
- (9) FIG. 8 is a block diagram of an embodiment of the wire feeder of FIG. 2 including a shielding gas flow control system for conveying shielding gas with the welding wire;
- (10) FIG. 9 is a block diagram of an embodiment of the wire feeder of FIG. 2 showing the removal of certain components from the wire feeder;
- (11) FIG. 10 is a perspective view of an embodiment of an attachment mechanism used to attach a flexible conduit of the wire feeder of FIG. 2 to a welding torch of a separate welding system;
- (12) FIG. 11 is a diagrammatical representation of the wire feeder of FIG. 2 including a return lead extending from an enclosure of the wire feeder to a work piece; and
- (13) FIG. 12 is a perspective view of an embodiment of the wire feeder of FIG. 2 attached to an arm of a user via a harness system.

DETAILED DESCRIPTION

(14) Present embodiments are directed to systems and methods for providing a portable welding wire feeder that is powered at least in part by a battery. The wire feeder includes, among other things, a spool for providing a continuous supply of welding wire, a wire drive system for conveying the welding wire from the spool through a flexible conduit extending from the wire feeder, a motor for powering the wire drive system, and a battery receptacle for receiving the battery. The battery may be removable from an enclosure of the wire feeder, and the battery may be rechargeable. The wire feeder may also include a harness system for attaching the wire feeder to the body of a user, allowing the user to move around while welding using the continuous wire feed from the wire feeder. The harness system and battery operation may increase portability of the wire feeder, allowing greater flexibility for the user.

(15) Turning now to the figures, FIG. 1 illustrates a tungsten inert gas (TIG) welding system 10 including a power supply 12, a shielding gas source 14, a cooling system 16, a wire feeder 18, and a welding torch 20. In the illustrated embodiment, the power supply 12 provides power to the welding torch 20 via a supply conduit 22. The power supply 12 may supply a direct current (DC) or alternating current (AC) to the welding torch 20 depending on the desired application. For example,

an AC current may be suited for welding aluminum or magnesium, and a DC current may be suited for welding stainless steels, nickel, or titanium. In addition to matching the current to the material selection, the output of the power supply **12** may be varied to obtain desired weld characteristics. For example, a low AC frequency (e.g., 60 Hz) current may generate a wide arc with shallow penetration of a work piece **24**, while a high AC frequency (e.g., 200 Hz) current may generate a focused arc with deeper penetration into the work piece **24**.

(16) In addition to the frequency of the current, the power supply **12** may vary the amperage of the current output to the welding torch **20**. The setting for the amperage output by the power supply **12** may be adjusted by setting a knob or button on the power supply **12**, or may be set by a remote control **26**. For example, a welding system **10** may include a foot pedal remote control **26** that allows the operator to make current adjustments during welding by either holding down the foot pedal or feathering the foot pedal remote control **26** to vary the amperage. The remote control **26** may also include a finger tip trigger control, audible command, or other form of input to signal the power supply **12** to output a corresponding current.

(17) In addition, the welding torch **20** may be supplied with a shielding gas from the shielding gas source **14**. In general, the shielding gas may be supplied to the welding torch **20** and expelled from the welding torch **20** at the location of the weld. The shielding gas may be expelled immediately prior to striking the welding arc, throughout welding, and/or until shortly after the welding arc is extinguished. The shielding gas protects the welding area from atmospheric gases such as nitrogen and oxygen, which can cause fusion defects, porosity, and weld metal embrittlement. The shielding gas may also transfer heat from the welding electrode to the metal and may help to start and maintain a stable arc.

(18) The shielding gas may be provided in a cylindrical container and delivered to the welding torch **20** via a conduit **28**, a gas valve **30**, and a supply conduit **32**. A regulator on the gas source **14** may allow an operator to limit the pressure of the gas delivered to the gas valve **30** to obtain a desired flow rate. Further, the gas valve **30** may provide for stopping and starting the flow of the shielding gas to the welding torch **20** in coordination with other welding operations.

(19) The welding system **10** may be provided with a cooling system **16** to reduce heat build-up in the welding torch **20**. The cooling system **16** may take various forms including gas cooled and liquid cooled systems. The cooling system **16** may provide for circulation of the coolant via a coolant supply conduit **34**, the supply conduit **22**, and a coolant return conduit **36**. The cooling system **16** may be powered from the power supply **12** via a coolant system power cord **38**.

(20) In general, the welding system **10** may provide for current flow via a work piece **24** to the power supply **12**. For example, as depicted in FIG. 1, the welding system **10** may include a cable **40** that is secured to the work piece **24** via a work clamp **42**. In this configuration, the current provided by the power supply **12** flows through the supply conduit **22** to the welding torch **20**, flows across an arc from an electrode **44** to the work piece **24**, and returns to the power supply **12** via the work clamp **42** and the cable **40**.

(21) The welding system **10** includes a wire feeder **18** that is powered at least partially by a battery, making the wire feeder **18** relatively portable. The wire feeder **18** may provide a continuous feed of welding wire **46** for use during the welding operation. This welding wire **46** is delivered toward the work piece **24** and deposited into the weld forming on the work piece **24** after being heated by the welding arc. Continuously fed welding wire **46** may offer an improvement to typically used filler material rods, as described above, that are fed manually to the weld, as the continuous feed reduces an amount of wasted filler material (e.g., a stub at the end of a filler rod) and an amount of time spent discarding and replacing the filler rod. In addition, the wire feeder **18**, which is portable and capable of being decoupled from other components of the welding system **10**, may offer an unprecedented degree of flexibility for the welding operator.

(22) Although the wire feeder **18** is at least partly battery powered, a power connection **48** may extend from the wire feeder **18** to receive additional power from the power supply **12**, or another

outside power source. In certain embodiments, power received by the wire feeder **18** through the power connection **48** may be used to recharge a battery within the battery powered wire feeder system **18**, or to supply a portion of the power used to operate the wire feeder **18**. In certain embodiments, other connections may couple the wire feeder **18** with the power supply **12** or other components of the welding system **10** for communicating signals to control a wire feed speed for the wire feeder **18**. As previously mentioned, the wire feeder **18** is at least partially battery operated. Because the wire feeder **18** is entirely battery operated, it may provide a portable, compact, and cordless system for feeding wire continuously to a weld site.

(23) It should be noted that the welding system **10** illustrated in FIG. **1** is specific to tungsten inert gas (TIG) welding applications. However, the portable wire feeder **18**, described in detail below, may be used in a variety of other welding contexts including, but not limited to, gas welding (e.g., oxyacetylene gas welding) and plasma welding. Similar to TIG welding, these welding applications generally involve a user manually feeding a filler material rod to the weld location with one hand while positioning the welding torch **20** with the other hand. Indeed, any welding application that typically employs consumable rods for depositing filler material into the weld may benefit from the wire feeder **18** described herein.

(24) FIG. **2** is a diagrammatical representation of an embodiment of certain components of the wire feeder **18** of FIG. **1**. In the illustrated embodiment, the wire feeder **18** includes, among other things, an enclosure **60**, a welding wire spool **62**, a welding wire drive system **64**, a motor **66**, and a battery **68**. The enclosure **60** may include a spool mount (e.g., a hub) **72** configured to receive the spool **62**, wire drive **64**, motor **66**, and a battery receptacle **70** configured to receive the battery **68**. Indeed, the battery **68**, which may be a rechargeable battery, may be removed from the battery receptacle **70** of the enclosure **60** when recharging or replacement of the battery **68** is desired. Similarly, the spool **62** may also be removed from the enclosure **60** and replaced when the welding wire **46** of the spool **62** has been consumed. The spool **62**, wire drive **64**, motor **66**, and battery **68** facilitate the continuous feeding of welding wire **46** through a flexible conduit **78** extending from the enclosure **60**. An operator may grasp the flexible conduit **78** at an end distal to the enclosure **60** in order to manipulate the welding wire **46** toward the arc formed between the electrode **44** and the work piece **24**.

(25) The welding wire **46** is supplied by the welding wire spool **62**, which may be mounted on the spool mount **72** within the enclosure **60**. It may be desirable for the size, weight, and type of welding wire **46** on the spool **62** to conform to an industry standard. For example, the spool **62** may have a diameter of approximately four inches in order to maintain a weight of the wire feeder **18** that is light enough to be portable. Such welding wire spools **62** generally weigh approximately one pound for aluminum welding wire or approximately two pounds for steel, stainless steel, or nickel welding wire **46**. The wire feeder **18** described herein may accommodate spools **62** of larger standard sizes as well, or may accommodate a custom sized spool **62** designed specifically for the wire feeder **18**.

(26) The illustrated welding wire drive system **64** includes a pair of feed wheels **74** and **76** that are configured to feed the welding wire **46** from the spool **62** through the flexible conduit **78** attached to the enclosure **60**. The first feed wheel **74** may be an idler wheel configured to rotate relative to the enclosure **60** as the welding wire **46** moves, while the second feed wheel **76** may be a drive wheel configured to be turned by the motor **66** in order to urge the welding wire **46** toward the flexible conduit **78**. In accordance with other embodiments of the wire feeder **18**, the welding wire drive system **64** may include two or more pairs of feed wheels driven by the motor **66** in order to feed the welding wire **46**.

(27) The motor **66** may include, among other things, a motor drive **80** and control circuitry **82** for controlling operation of the motor drive **80**, as discussed in detail below. The motor **66** is powered at least in part by the battery **68** located in the battery receptacle **70** of the enclosure **60**. In certain embodiments, the motor **66** may receive supplemental power from the power supply **12** of the

welding system **10**, or any other external power supply, via the optional power connection **48**. Indeed, the power connection **48** may receive power from the external power source (e.g., power supply **12**) for recharging the battery **68** or providing additional power to drive the motor **66**. Other embodiments may include a quick recharge feature for use when the battery **68** is rechargeable. Such a feature may include charging circuitry in communication with the power connection **48** to facilitate recharging of the battery **68** while the battery **68** is in the enclosure **60** of the wire feeder **18**. In the illustrated embodiment, the enclosure **60** does not include the power connection **48**, making the wire feeder system **18** entirely battery operated and cordless. This may increase the portability and flexibility of operation of the wire feeder **18**.

(28) FIG. **3** is a block diagram of an embodiment of certain components of the wire feeder **18**, including various components of the control circuitry **82** for controlling operation of the wire feeder **18**. As illustrated, the control circuitry **82** may include a motor drive circuit **94**, a processor **96**, a memory **98**, interface circuitry **100**, and operator interface circuitry **102**. These components of the control circuitry **82** may be located within the motor **66**, as shown in FIG. **2**, or in one or more different locations within the enclosure **60** of the wire feeder **18**. Certain circuitry components may communicate with components of the welding system **10** located generally outside the enclosure **60** in order to control a wire feed speed of the welding wire **46** exiting the wire feeder **18** toward the welding application.

(29) The motor drive circuit **94** communicates with the processor **96** in order to operate the motor drive **80** at an appropriate speed for feeding the welding wire **46** supplied by the spool **62** through the flexible conduit **78**. The processor **96** may determine the appropriate speed based on signals from the interface circuitry **100** and operator interface circuitry **102**. In addition, the processor **96** may be operably coupled to the memory **98** in order to execute instructions for determining the appropriate wire feed speed based on these signals.

(30) The interface circuitry **100** may communicate power supplied by the battery **68** to the processor **96** to power operations of both the processor **96** and the motor drive **80**. The interface circuitry **100** also may convey additional power from certain components of the welding system **10** (e.g., the power supply **12** via the power connection **48**) for operating the motor drive **80**. Further, the interface circuitry **100** may transmit signals related to various operating parameters of the welding system **10** (e.g., current output of the power supply **12**) in order to control wire feed speed. For example, when the power supply **12** provides or adjusts a current to the welding torch **20**, a corresponding signal may be communicated to the processor **96** via the interface circuitry **100**. Accordingly, a signal may be sent from the processor **96** to the motor drive circuit **94** for turning on or adjusting the speed of the motor drive **80**. Signals from components of the welding system **10** that are separate from the wire feeder **18** may be communicated wirelessly or through a cable that plugs into the welding system **10** to activate the welding wire drive system **64**. Wireless communication between the welding system **10** and the wire feeder **18** may allow a certain ease of manipulation of the wire feeder **18** relative to other welding system components, further increasing portability of the wire feeder **18**.

(31) The operator interface circuitry **102** may include user input devices of the wire feeder **18** for manually adjusting the wire feed speed. For example, the operator interface circuitry **102** may be linked to a control panel located on an outer surface of the enclosure **60**. In other embodiments, the operator interface circuitry **102** may communicate signals from a trigger **104** positioned along the flexible conduit **78**. Indeed, an operator may manipulate the welding torch **20** in one hand and the flexible conduit **78** for conveying the welding wire **46** in the other hand, activating the welding wire drive system **64** of the wire feeder **18** based on depression of the trigger **104** while continuously welding.

(32) FIG. **4** is a perspective view of an embodiment of the wire feeder **18** of FIG. **2** having the enclosure **60** attached to a torso **116** of a user **118** via a harness system **120** such that welding wire **46** is conveyed to the left side of the user **118**. The harness system **120** of the wire feeder **18** may be

used for attaching the enclosure **60** to a body of the user **118**, which may include, among other things, the torso **116**. In the illustrated embodiment, the harness system **120** includes straps through which the arms of the user **118** may be inserted. In this way, the portable wire feeder **18** may be worn as a backpack, secured to the back of the user's torso **116** so that the user **118** may maneuver the welding torch **20** in a right hand **122** and the flexible conduit **78** in a left hand **124**. The previously discussed trigger **104** may be located along an end portion of the flexible conduit **78** such that the user **118** may control the speed of the welding wire **46** exiting the wire feeder **18** during a welding process. Since the wire feeder **18** is powered by the battery **68** and worn around the torso **116** of the user **118**, the user **118** may manipulate both the welding torch **20** and the welding wire **46** exiting the flexible conduit **78** with relatively greater ease than possible using a larger bench-style wire feeder.

(33) It should be noted that other arrangements of components within the enclosure **60** of the wire feeder **18** may be possible, and other arrangements of the harness system **120** may be used to secure the enclosure **60** to the torso **116** of the user **118**. Certain arrangements of the wire feeder **18** may be worn on the front or the back of the user **118**, and some embodiments allow the user **118** to manipulate the welding wire **46** using either the right hand **122** or the left hand **124**. For example, FIG. 5 shows the enclosure **60** attached to the user's back in order to convey the welding wire **46** through the flexible conduit **78** to the right hand **122** of the user **118**. This arrangement may be especially useful for a left-handed user **118**. Indeed, the wire feeder **18** of FIG. 5 may be the wire feeder **18** of FIG. 4 rotated 180 degrees. In other words, the wire feeder **18** and associated harness **120** may be easily reversible in certain embodiments. For example, the harness system **120** may be removed from the user **118**, the entire wire feeder **18** turned 180 degrees, and the harness system **120** replaced on the user **118** to switch between these different arrangements.

(34) FIG. 6 is a perspective view of an embodiment of the wire feeder **18** of FIG. 2 attached to the back of the user **118** via the harness system **120** such that welding wire **46** is conveyed upward from the enclosure **60** through the flexible conduit **78**. This configuration of the harness system **120** relative to the enclosure **60** and components internal to the enclosure **60** may allow the user **118** to manipulate the flexible conduit **78** using the right hand **122** or the left hand **124** without removing the enclosure **60** from the user's torso **116**. It should be noted that the battery **68**, the motor **66**, the welding wire drive system **64**, and the spool **62** may be arranged in the enclosure **60** in any suitable configuration for directing the welding wire **46** from the enclosure **60** through the flexible conduit **78** in an upward direction.

(35) FIG. 7 illustrates the wire feeder **18** having the enclosure **60** attached to a front side of the user's torso **116**. The wire feeder **18** in the illustrated embodiment may be the same wire feeder as shown in either or both of FIGS. 4 and 5, attached via the harness system **120** to the user's front instead of the user's back. This may allow the user **118** to manipulate the welding wire **46** with the right hand **122** while maintaining the orientation of internal components of the wire feeder **18** as shown in FIG. 4.

(36) FIG. 8 is a block diagram of an embodiment of the wire feeder **18** including a shielding gas flow control system **136** for conveying shielding gas **138** through the flexible conduit **78** along with the welding wire **46**. This may be useful for routing the shielding gas **138** through the flexible conduit **78** to shield the welding wire **46** from contaminants as the welding wire **46** is melted onto the work piece **24**. This shielding gas flow control system **136** may provide the shielding gas **138** to the weld in addition to or in lieu of shielding gas supplied through the welding torch **20**. In the illustrated embodiment, the enclosure **60** includes a gas cylinder receptacle **140** for receiving a shielding gas cylinder **142**, and the shielding gas flow control system **136** is designed to receive the shielding gas **138** from the shielding gas cylinder **142** when the shielding gas cylinder **142** is inserted into the shielding gas cylinder receptacle **140**. In certain embodiments, the shielding gas flow control system **136** may control the amount of shielding gas **138** delivered through the flexible conduit **78** (e.g., using flow control valves) based on the speed of the motor drive **80**, as determined

by the control circuitry **82** of the wire feeder **18**. The shielding gas cylinder **142** may be removed from the shielding gas cylinder receptacle **140** and replaced with another shielding gas cylinder **142** when the shielding gas **138** is depleted. The shielding gas **138** may be a similar or different inert gas (e.g., argon) than the shielding gas provided to the welding torch **20** by the gas source **14**. Furthermore, the shielding gas flow control system **136** may receive the shielding gas **138** from the gas source **14** in order to deliver the shielding gas **138** through the flexible conduit **78**.

(37) FIG. **9** is a block diagram of an embodiment of the wire feeder **18** showing the removal of certain components from the enclosure **60**. It may be desirable to remove such components as the welding wire spool **62** and/or the battery **68** from the enclosure **60** of the portable wire feeder **18**. As the welding wire drive system **64** conveys the welding wire **46** from the spool **62** through the flexible conduit **78**, the spool **62** may eventually empty of the welding wire **46**. At this point, a new welding wire spool **62** may be inserted into the enclosure **60** to supply the welding wire **46**. To accommodate this, a spool receptacle **150** may be located in the enclosure **60** to allow for insertion and/or removal of the spool **62**.

(38) Likewise, it may be desirable for the battery **68** to be removed from the enclosure **60** for battery replacement or recharging (if the battery **68** is rechargeable). The illustrated embodiment shows both the battery **68** and the spool **62** being removed from their respective receptacles **70** and **150**. Arrows **152** and **154** indicate downward directions in which the battery **68** and the spool **62**, respectively, may be lowered from the enclosure **60**. The enclosure **60** may include one or more covers that are hinged or removable to facilitate removal of the battery **68**, the spool **62**, and other components (e.g., the shielding gas cylinder **142** if present) from the enclosure **60**. In certain embodiments, such covers may each feature a quick release mechanism, such as a snap fit feature, to allow the user **118** or someone else to easily open the cover(s) to remove the spool **62** and/or the battery **68**. It should be noted that the enclosure **60** may be arranged to allow removal of these internal components in other directions; however, removal through the bottom of the enclosure **60** may enable the user **118** to remain wearing the wire feeder **18** on the user's torso **116** while someone else replaces the component(s). In addition, the process of removing the spool **62** from the enclosure **60** may include additional steps such as removing the spool **62** from the spool mount **72** upon which the spool **62** is situated during wire feeder operation.

(39) FIG. **10** is a perspective view of an embodiment of an attachment mechanism **166** used to attach the flexible conduit **78** of the wire feeder **18** to the welding torch **20**. The attachment mechanism **166** may be part of the flexible conduit **78**, an integral component of the welding torch **20**, or a separate component that may be selectively attached to both the flexible conduit **78** and the welding torch **20**. In the illustrated embodiment, the welding torch **20** includes a torch handle **168**, a torch nozzle **170**, and a back cap **172**. The illustrated attachment mechanism **166** is attached to the welding torch **20** near the handle **168**, between the nozzle **170** and back cap **172**, but in other embodiments, the attachment mechanism **166** may attach at different points along the welding torch **20**.

(40) In some embodiments, the attachment mechanism **166** may include a quick release feature that allows the flexible conduit **78** to be easily engaged or disengaged with the welding torch **20** when desired. During welding operations, an operator may desire to reorient the welding torch **20** relative to the work piece **24** in order to access the weld site. To accommodate this reorientation of the welding torch **20**, it may be desirable for the attachment mechanism **166** to allow a corresponding adjustment of the orientation of the flexible conduit **78** relative to the welding torch **20**. For example, the attachment mechanism **166** may rotate relative to an axis **174** of the welding torch **20** as it attaches the flexible conduit **78** to the welding torch **20**. This would allow an angle **176** between the flexible conduit **78** and the welding torch **20** to be adjusted so that the flexible conduit **78** remains attached to the welding torch **20** throughout the welding process, even as the welding torch **20** is repositioned relative to the work piece **24**.

(41) FIG. **11** is a diagrammatical representation of the wire feeder **18** feeding the welding wire **46**

through the flexible conduit **78** in a hot wire mode. That is, the wire feeder **18** supplies a charge carrying welding wire **46** to the work piece **24**. The current flowing through the welding wire **46** heats the wire to a relatively higher temperature than possible without the current, allowing the welding wire **46** to be deposited more quickly on the work piece **24** to make the welding process more time efficient. In order to facilitate the current flowing through the welding wire **46**, the wire feeder **18** may include a return lead **186** extending from the enclosure **60** to the work piece **24** via a cable **188**. The wire feeder **18** may include a relatively small power source connected to both the welding wire **46** and the return cable **188**. This power source may be the same battery **68** used to power the motor **66** for feeding the welding wire **46**, or a separate power source. As the welding wire **46** is brought into contact with the work piece **24**, a loop closes between the cable **188**, work piece **24**, welding wire **46**, and power source, allowing the current to flow through and heat the welding wire **46**. Thus, the wire feeder **18** may support hot wire welding or cold wire welding. (42) It should be noted that the return lead **186** may be a clamp similar to the work clamp **42** of FIG. **1**, extending from the enclosure **60** via the cable **188** and fixed to the work piece **24** throughout the welding operation. In other embodiments, however, the return lead **186** extending from the enclosure **60** may be selectively brought into contact with the work piece **24** during welding. For example, the return lead **186** may be attached to a glove worn by the user **118** in order to allow the user **118** greater flexibility in positioning a point of contact between the return lead **186** and the work piece **24**.

(43) FIG. **12** is a perspective view of an embodiment of the wire feeder **18** of FIG. **2** attached to an arm **190** of the user **118** via the harness system **120**. As previously mentioned, the harness system **120** may be used for attaching the enclosure **60** of the wire feeder **18** to a body, which may include the torso **116**, the arm **190**, or other body parts of the user **118**. Attaching the wire feeder **18** to the arm **190** may be beneficial for supplying a continuous feed of the welding wire **46** toward the hand (e.g., the right hand **122**) of the arm **190** due to the close proximity of the enclosure **60** to the right hand **122**. As a result, the wire feeder **18** may provide a continuous feed of the welding wire **46** for use in a welding operation, while enabling the user **118** to manipulate the welding wire **24** and other welding equipment with an unprecedented degree of flexibility.

(44) While only certain features of the invention have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.

Claims

1. A portable, battery powered wire feeder for tungsten inert gas (TIG) welding, comprising: an enclosure, wherein the enclosure comprises: a hub for receiving a welding wire spool; a welding wire drive system for receiving welding wire from the welding wire spool and feeding the welding wire through a flexible conduit attached to the enclosure, the flexible conduit to deliver the welding wire to a workpiece, the welding wire to be melted by a tungsten inert gas (TIG) welding torch at the workpiece, the TIG welding torch comprising a non-consumable tungsten electrode that is independent from the welding wire from the spool, wherein an entirety of the flexible conduit is independently movable and separate from a cable supplying power and gas to the TIG welding torch that consumes the welding wire as the welding wire is fed through the flexible conduit during a TIG welding operation; a motor for driving the welding wire drive system, wherein the motor is powered at least in part by a battery; and a battery receptacle for receiving the battery.
2. The wire feeder of claim 1, comprising a harness system for attaching the enclosure to a body of a welder.
3. The wire feeder of claim 1, wherein the welding wire drive system is activated based on depression of a trigger.

4. The wire feeder of claim 1, wherein the welding wire drive system is activated based on a signal received from a separate welding system.
 5. The wire feeder of claim 4, wherein the signal is received from the welding system wirelessly.
 6. The wire feeder of claim 1, wherein the enclosure comprises a power connection for receiving power from an external power source.
 7. The wire feeder of claim 1, wherein the enclosure comprises a shielding gas flow control system for receiving shielding gas and delivering the shielding gas through the flexible conduit.
 8. The wire feeder of claim 7, wherein the enclosure comprises a shielding gas cylinder receptacle for receiving a shielding gas cylinder, wherein the shielding gas flow control system receives the shielding gas from the shielding gas cylinder when the shielding gas cylinder is inserted into the shielding gas cylinder receptacle.
 9. The wire feeder of claim 1, comprising an attachment mechanism for attaching the flexible conduit to the TIG welding torch, wherein the TIG welding torch is configured to receive power from a separate welding system.
 10. The wire feeder of claim 1, wherein the welding wire spool is removable from the enclosure and replaceable.
 11. A portable, cordless wire feeder system for tungsten inert gas (TIG) welding comprising: a welding wire feeder, comprising: a hub for receiving a welding wire spool; a welding wire drive system for receiving welding wire from the welding wire spool and feeding the welding wire through a flexible conduit and delivering the welding wire to a workpiece, the welding wire to be melted by a TIG welding torch at the workpiece, the TIG welding torch comprising a non-consumable tungsten electrode that is independent from the welding wire from the spool; a motor for driving the welding wire drive system, wherein the motor is powered at least in part by a battery; a trigger to activate the motor to drive the welding wire drive system during a TIG welding operation, the trigger being independent of a trigger for the TIG welding torch; a battery receptacle for receiving the battery; and a harness system for attaching the welding wire feeder to a body of a user.
 12. The portable, cordless wire feeder system of claim 11, wherein the welding wire drive system is activated based on depression of a trigger.
 13. The portable, cordless wire feeder system of claim 11, wherein the welding wire drive system is activated based on a signal received from a separate welding system.
 14. The portable, cordless wire feeder system of claim 13, wherein the signal is received from the welding system wirelessly.
 15. The portable, cordless wire feeder system of claim 11, wherein the welding wire feeder comprises a shielding gas flow control system for receiving shielding gas and delivering the shielding gas through the flexible conduit.
 16. The portable, cordless wire feeder system of claim 11, wherein the welding wire feeder comprises a shielding gas cylinder receptacle for receiving a shielding gas cylinder, wherein the shielding gas flow control system receives the shielding gas from the shielding gas cylinder when the shielding gas cylinder is inserted into the shielding gas cylinder receptacle.
 17. The portable, cordless wire feeder system of claim 11, comprising an attachment mechanism for attaching the flexible conduit to a welding torch of a separate welding system.
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